MAUVE-MUSE Paper II: environmental suppression of star formation and the changing excitation of ionised gas

Expanded literature assessment and revised Introduction | 27 October 2026

Prepared for Rongjun Huang. This report replaces Parts A, B, and H of the 26 October narrative report. Part A develops the argument against a substantially expanded literature base; Part B is continuous manuscript prose; Part H provides the annotated bibliography. Parts C-G of the earlier report, the source notebooks, and their saved results are unchanged. Where the expanded literature changes an interpretation in those earlier parts, the correction is stated here.

Evidence and scope. This is a literature and manuscript revision, not a new measurement of the MAUVE data. MAUVE statements refer to the user-described findings and the source/saved-output assessment documented on 26 October. No notebook or FITS pipeline was executed. The literature search was refreshed through 27 October 2026, including primary author manuscripts, journal articles, and the VESTIGE, VERTICO, and GASP publication lists. Strong claims were checked against the relevant results or discussion where available; an abstract-level check is not represented as a complete paper audit. The Introduction follows the progression, author-year citation density, and distinction between observation and interpretation in the local author version of Paper I, Huang et al. (2026).

Draft Introduction: 2915 words | 111 annotated references

Part A develops the scientific argument and terminology. Part B provides the expanded manuscript Introduction. Part H collects the sources and their specific roles.

Contents

This document replaces Parts A, B and H of the 26 October report. The section labels below refer to this report; the seven-section manuscript progression is described in A10.

A. Revised scientific argument and literature assessment . . . 3

A1. The central argument after the expanded search . . . 3

A2. What the expanded search covers . . . 3

A3. The measurements that must not be conflated . . . 4

A4. Do previous studies find that H II regions are fainter, as well as fewer? . . . 5

A5. VERTICO: suppressed SF is not synonymous with exhausted molecular gas . . . 5

A6. GASP, MaNGA, and other resolved precedents . . . 6

A7. What can be claimed, and what the nearest precedents already establish . . . 8

A8. Consequences for the physical discussion . . . 9

A9. Terminology: should the paper use 'residual emission'? . . . 9

A10. The resulting manuscript progression . . . 10

B. Revised manuscript Section 1: Introduction . . . 12

H. Expanded annotated bibliography . . . 17

H1. Environment, quenching, and the broad physical framework . . . 17

H2. Virgo histories, VESTIGE, and H II-region properties . . . 18

H3. Molecular gas, VERTICO, and the gas-SF connection . . . 19

H4. Resolved SF relations, completeness, and environmental IFS studies . . . 20

H5. Local enhancement, late-stage suppression, and simulation comparisons . . . 22

H6. DIG, LIER emission, and excitation in stripped systems . . . 23

H7. Line diagnostics, density, and kinematic interpretation . . . 24

H8. MAUVE context, Paper I, and local feedback . . . 25

A. Revised scientific argument and literature assessment

A1. The central argument after the expanded search

The paper should ask: **How are the intensity and spatial coverage of star formation connected to the distribution and excitation of detectable ionised gas across the adopted Virgo infall-stage populations?** This question accommodates the full sequence of proposed results: the rSFMS comparison; the decline in conditional SF intensity; the limited cases of local excess; the changing SF/NSF/ND census; and the weaker but differently excited emission outside the adopted SF selection.

The literature now supports a more specific starting point than yesterday's Introduction. Environmental studies have already established reduced SF, truncated discs, changes in the luminosities of H II regions, reduced molecular-gas content and efficiency, extended low-ionisation emission, and links between stripping state and excitation. The opportunity for Paper II is to measure how these manifestations are connected **within one spatially resolved census**, retaining the distinction between the fraction of the usable footprint forming stars and the intensity of SF within that selected component.

The proposed working title is:

> ***MAUVE-MUSE: star formation suppression and changing ionised-gas excitation along the Virgo infall sequence***

Here, 'sequence' refers to the adopted pre-peak, close-to-peak, and post-peak populations. The text must explain that these are different galaxies, rather than repeated observations of an individual galaxy or nebula. A more conservative title, if the stage calibration remains a substantial uncertainty, is **MAUVE-MUSE: the spatial connection between star formation suppression and ionised-gas excitation in Virgo galaxies**. Neither title requires 'residual emission', a universal density decline, or a uniquely established HOLMES contribution.

A2. What the expanded search covers

The search followed six connected questions: the global environmental dependence of SF; the spatial suppression of SF in Virgo and other clusters; the luminosity and surface brightness of the remaining SF component; the distinction between molecular-gas supply and SF efficiency; the spatial census of weak/non-H II-like emission; and competing explanations for its excitation. Search terms combined the survey names with 'H II luminosity', 'surface brightness', 'resolved main sequence', 'truncation', 'quenching', 'diffuse ionised gas', 'LIER', 'equivalent width', 'gas density', and 'star formation efficiency'. Reference lists were followed from the closest studies to earlier precedents and newer results.

The enlarged coverage includes foundational environmental work; VESTIGE imaging and individual-region studies through 2026; VIVA histories; VERTICO I-VII; GASP resolved SF, clump, DIG, quenching-history, and full-sample studies; MaNGA and SAMI environmental and excitation analyses; PHANGS-MUSE and MAD comparisons; and relevant ionisation models. The bibliography also retains selected discussion references that do not all belong in the Introduction. Survey publication pages were discovery tools: conclusions below are attached to individual papers, because a survey name does not specify a selection, tracer, or estimator.

This is a comprehensive targeted narrative review, rather than a formally registered systematic review or a demonstration of absolute bibliographic completeness. In particular, a negative search result cannot

establish that an observational combination has never been published. The appropriate contribution statement is therefore a defined extension of the nearest precedents, without an unrestricted 'first' claim.

A3. The measurements that must not be conflated

The missing distinction in the previous draft was not merely between detected and undetected SF. Several different observables are described informally as 'fading', and they answer different questions.

Observable	What it measures	Relevance to MAUVE
Number of segmented H II regions above a luminosity limit	The abundance of identified nebulae, with a catalogue completeness boundary	Relevant to the loss of SF sites, but not identical to the area fraction of SF spaxels
Integrated Hα luminosity of one region	Its total recombination luminosity; depends on size, stellar content, age, attenuation, leakage, and blending	A lower value does not automatically mean lower SFR per unit area
Brightest-region luminosity or a luminosity function	The bright extreme or distribution of a regional population	Depends on the number of sampled regions and on the host quantity held fixed
Hα luminosity per projected region area	Surface brightness of a segmented nebula	Closer to intensity, but the segmentation and PSF remain relevant
SFR surface density in SF-selected spaxels at fixed stellar surface density	The strength of recent SF conditional on passing the adopted selection	The most direct description of the stage-separated MAUVE rSFMS
SF area fraction	The fraction of a specified usable footprint assigned to SF	Describes coverage, rather than the brightness of the selected component
Molecular SF efficiency, SFR/molecular-gas mass	The rate of forming stars per unit molecular reservoir	Requires gas information; a suppressed rSFMS alone does not measure it
DIG Hα flux fraction	The diffuse contribution to the total Hα luminosity under a particular decomposition	Can increase while the diffuse luminosity decreases; is not an NSF area fraction

For a consistently weighted footprint, its mean SF-attributed SFR surface density can be expressed as the SF coverage multiplied by the conditional mean SF intensity. With equal-galaxy averaging, the conditional intensity associated with this identity must use the corresponding occupancy weights; it is not generally the arithmetic mean of the per-galaxy conditional intensities. This decomposition gives Section 4 a physical purpose: it distinguishes a smaller SF footprint from weaker SF within that footprint. It should not be relabelled as a catalogue of individual nebular luminosities.

The manuscript-safe description of the present result is therefore **'lower SFR surface density among SF-selected spaxels at comparable local stellar surface density in the later-stage populations'**. 'Fainter surviving H II regions' is acceptable shorthand only after explaining the observational meaning. 'Individual H II regions fade along infall' would require a temporal inference that the current population comparison does not directly provide.

A4. Do previous studies find that H II regions are fainter, as well as fewer?

Yes, there is directly relevant precedent, but it does not support a universal fading of every surviving region. The earlier report cited the regional-abundance result while omitting the more direct luminosity literature. Boselli et al. (2020; VESTIGE VI) establishes the deficit of outer H II regions on approximately 50 pc scales. It should not be used alone to substantiate a decline in conditional SF intensity. Boselli et al. (2025; VESTIGE XVII) supplies the regional luminosity, size, and density baseline in H I-normal systems.

The crucial omitted comparison is Boselli et al. (2026; VESTIGE XXI): the brightest/first-three-ranked regions are less luminous at fixed host stellar properties, but have similar luminosities at fixed SFR. Perturbed systems have a steeper composite faint end and brighter characteristic luminosity, without a clear trend in individual-galaxy fit parameters. They also show more high-density regions and a larger diffuse Hα contribution. This is not uniform nebular dimming.

This result materially changes the priority assessment. Paper II cannot claim that environmentally affected galaxies were previously known only to contain fewer regions, with the intensity of the remaining SF entirely unexplored. Equally, it should not claim a one-to-one confirmation of its own rSFMS result: a host-mass-conditioned brightest-region statistic differs from a local-stellar-density-conditioned spaxel statistic. An extreme regional luminosity can respond to the number of draws from the luminosity distribution. Testing an intrinsic shift of the regional population therefore requires completeness and sampling controls, rather than only a comparison of the brightest object.

There are also clear precedents for retained or enhanced intensity. Koopmann & Kenney (2004a,b) established that truncated Virgo discs can retain approximately normal or enhanced inner SF, alongside other morphological classes (companion morphology study). Boselli et al. (2022; VESTIGE XII) finds compact star-forming discs in Virgo lenticulars with SF surface brightness comparable to late-type galaxies, despite low integrated SFRs associated with their small extent. This is an especially useful demonstration that a galaxy can move below the global SFMS without every surviving SF location becoming faint.

For enhancement on individual-region scales, Boselli et al. (2021; VESTIGE IX) connects bright, denser H II regions in IC 3476 to gas compression. In six strongly stripped GASP galaxies, Giunchi et al. (2023) finds Hα clumps brighter at a given size than comparison regions in ordinary local star-forming galaxies. That sample is a relevant counterexample to universal suppression, not evidence that all infalling galaxies should enhance their SF. H II luminosity distributions also vary with internal galactic environment and ionisation properties in PHANGS-MUSE (Santoro et al. 2022).

The unresolved question that MAUVE can address is consequently narrower and more useful: **after using the same SF selection and a common range of local stellar surface density, does the conditional intensity decline across the adopted stages, and how much of the overall decline is associated with reduced SF coverage?** A measured decline in both quantities would establish simultaneous changes in the abundance and strength of SF-selected emission. It would connect the earlier regional, radial, and gas-scaling results without claiming that every nebula undergoes the same evolution.

A5. VERTICO: suppressed SF is not synonymous with exhausted molecular gas

The VERTICO sequence belongs near the centre of the literature argument. Its relevance extends beyond a single citation to the rSFMS. The survey design is described by Brown et al. (2021), and the following studies separate several routes to reduced SF.

Study	Result needed for Paper II	Consequence for the interpretation
Zabel et al. (2022), VERTICO II	H I-deficient systems have steeper, truncated molecular-gas profiles, especially through loss of low-surface-density outer material	A model retaining all molecular gas is a restricted scenario
Jimenez-Donaire et al. (2023), VERTICO III	An approximately linear ensemble KS relation coexists with galaxy-to-galaxy differences and lower molecular efficiency in H I-deficient systems	A familiar ensemble relation does not imply environmental invariance
Villanueva et al. (2022), VERTICO IV	Within the effective radius, increasingly perturbed H I is accompanied by lower molecular SF efficiency	Inner suppression can occur while a substantial molecular reservoir persists
Watts et al. (2023), VERTICO V	Environmental effects lower the surface density of retained atomic gas and, in strongly deficient galaxies, molecular gas	The retained inner ISM is not necessarily unaffected
Roberts et al. (2023), VERTICO VI	Leading halves of H I-tailed discs show excess molecular gas and SF, with an approximately unchanged depletion time	Local enhancement can reflect more fuel without enhanced efficiency
Brown et al. (2023), VERTICO VII	At 720 pc, resolved SF suppression is associated with both reduced molecular content and reduced efficiency	This is a close precedent for weaker SF at fixed local stellar density

These papers make two aspects of the proposed MAUVE interpretation more compelling. First, the remaining inner ionised emission should be considered in the context of a transformed, potentially depleted or inefficient cold-gas reservoir. Second, the NGC 4654 excess should be framed as a local departure within a population that can be suppressed overall. Neither interpretation requires the gas content or the efficiency to respond in the same way everywhere. Earlier Virgo measurements already found that enhanced molecular efficiency is not a universal response to ram pressure (Vollmer et al. 2012).

A6. GASP, MaNGA, and other resolved precedents

GASP establishes both enhancement and later suppression. The integrated enhancement study of Vulcani et al. (2018), the resolved SFR-mass analysis of Vulcani et al. (2020b), and the HST clump study cited above belong together. They describe selected stripping populations, with different physical scales and tracers. They do not require a population-wide enhancement in a sample extending to strongly suppressed stages. The abruptly quenched systems studied by Vulcani et al. (2020a) and the full MUSE sample of Poggianti et al. (2025) supply the necessary late-stage context. The latter also discusses selection incompleteness and galaxies with centrally restricted gas discs.

The wider literature does not show a universal enhancement. Foster et al. (2025) finds no systematic asymmetric SF enhancement in a large UNIONS satellite sample, while Goller et al. (2023) finds no population-wide enhancement in simulated TNG50 jellyfish despite bursts along individual histories. Moretti et al. (2026) reports normal-to-low efficiencies in four GASP jellyfish, with substantial dependence on location and gas density. These comparisons favour a stage-, selection-, and scale-dependent account rather than a single expected sign of the environmental response.

The GASP DIG papers directly constrain the proposed novelty. Tomicic et al. (2021a) compares stripped and control galaxies and finds no clear difference in their inferred DIG flux fractions. A larger MAUVE NSF area fraction would not contradict that result, because its denominator and classification differ. Tomicic et al. (2021b) links enhanced [O I]/Hα in diffuse tails to additional ionisation processes. Tomicic et al. (2024) associates UV/Hα discrepancies in DIG-dominated regions of late-stage systems with recently declining SF. That study also identifies Hα-emitting areas without a UV counterpart and with non-SF excitation. Thus, neither detectable gas after declining SF nor a changing relation between SF tracers is unprecedented.

There is also a particularly close conceptual antecedent outside GASP. Pedrini et al. (2022) relates interface-region excitation, SF activity, and gas-disc truncation to different stripping phases in A1367 and ESO 137-001. Its late-stage interpretation emphasises low SF and shocks at the interface. This is not the same geometry or estimator as MAUVE's inner-disc NSF census, but it rules out presenting the general connection between stripping stage and excitation as new. The relevant sections of that paper and Fossati et al. (2019) support mixed photoionisation and shocks; they should not be cited alone as proof of old-star ionisation in MAUVE discs.

MaNGA provides both environmental SF and weak-emission precedents. Spindler et al. (2018) finds lower satellite specific SFRs across radii; Lian et al. (2019) finds an environmental dependence of SF profiles particularly in low-mass satellites; and Bluck et al. (2020a,b) separates the predictors of regional SF from quenching and examines the radial patterns of satellite quenching (radial analysis). Some of these analyses supplement emission-line SFRs with continuum-based estimates in weak-line regions. Their 'quenched spaxels' therefore cannot be equated with MAUVE's Balmer non-detections or NSF selection.

Belfiore et al. (2016, 2017) already establishes extended low-ionisation emission, weak equivalent widths, and the resolved transition towards quiescence (transition study). Zhang et al. (2017) establishes the connection between low Hα surface brightness and enhanced forbidden-line ratios. Central LIER emission in a galaxy that forms stars further out is not, by itself, an outside-in stripping sequence; the environmental context must be supplied independently.

The SAMI results strengthen the comparison. Schaefer et al. (2017) and Medling et al. (2018) provide environmental and resolved-SF precedents. More directly, Cakir et al. (2026) compares unperturbed, asymmetric, and truncated cluster galaxies, finding stronger outer suppression in the more affected classes without a corresponding central enhancement. This is a close contemporary comparison for the proposed stage-dependent rSFMS, although at a coarser physical resolution.

A7. What can be claimed, and what the nearest precedents already establish

MAUVE finding or intended claim	Established context	Defensible addition in Paper II
The full resolved SF distribution overlaps a nearby-disc comparison	CALIFA, MAD, PHANGS, and GASP establish resolved SF relations with selection-dependent scatter	Test the comparison with explicitly aligned scale, footprint, and treatment of non-detections
Later stages have weaker conditional SF intensity	VESTIGE regional luminosities, VERTICO resolved suppression, and SAMI/MaNGA environmental profiles already bear on this question	Resolve the stage dependence at MAUVE scales and quantify it separately from coverage
Outer SF coverage declines	Virgo Hα imaging, VESTIGE, stellar-population studies, and GASP already establish truncation and outside-in suppression	Measure the accompanying SF/NSF/ND redistribution over the same usable footprint
Inner regions retain emission outside the SF selection	MaNGA LIERs, VESTIGE inner gas structures, GASP DIG studies, and prior MAUVE work provide precedents	Establish where the NSF area fraction changes with stage and local stellar density
NSF emission becomes fainter while diagnostic ratios rise	DIG surface-brightness sequences and environmental excitation differences are established	Compare absolute line surface brightness and ratios together within the stage-selected NSF population
Local enhancement can coexist with broad suppression	IC 3476, GASP clumps, VERTICO asymmetries, and NGC 4654 studies already show local responses	Quantify the NGC 4654 angular excess relative to a fixed internal SF reference
A declining young-star contribution reveals harder emission	HOLMES, leakage models, and Brown et al. (2026) already develop relevant interpretations	Test a restricted supply-decline model against the jointly measured spatial and spectroscopic trends

The closest MAUVE precedent must be explicit. Brown et al. (2026) already argues that suppressed SF exposes a larger DIG contribution, with a secondary shock-like component, in an early MAUVE sample. Paper II should build on that interpretation by resolving the stage and spatial dependence of the census, conditional SF intensity, and NSF line brightness. 'Environmental suppression unveils DIG' is therefore background to the proposed paper, rather than its standalone new conclusion.

For the PHANGS comparison, similar nominal spatial resolution does not establish equivalent measurements. The present notebook uses native spatial products rather than a demonstrated common-PSF 100 pc reconstruction; its upper SFR proxy includes NSF line emission and non-detection limits. This differs from the linear averaging and DIG treatment in Pessa et al. (2021). PHANGS also includes Virgo members and is not a strictly independent field-only control. The correct provisional statement is broad overlap under the stated comparison, with quantitative equivalence requiring matched estimators. MAD and VESTIGE additionally show that PHANGS is not the only relevant high-resolution survey, although it is an especially useful MUSE comparison.

A8. Consequences for the physical discussion

The outside-in pattern remains a strong organising observation, but its physical interpretation should use the gas morphology, stage information, and stellar constraints together. Morgan et al. (2024; VESTIGE XVI) shows that a starvation model incorporating a critical gas surface density can produce moderate SF-disc truncation. It discusses early starvation followed by stronger ram pressure. Accordingly, 'truncation excludes starvation' is too broad; the strength, timescale, and accompanying gas signatures determine the comparison. The more detailed NGC 4330 reconstruction of Fossati et al. (2018) illustrates how stellar information can constrain a rapid, radially structured decline.

The 20 October supply-decline model is useful as a controlled explanation, but it is not a complete environmental model. It suppresses replenishment while retaining a simplified molecular reservoir and prescribed conversion/depletion rates. VERTICO and Boselli et al. (2014) motivate alternatives involving direct molecular-gas loss or changing efficiency. Morgan et al. also supplies a close precedent for coupled atomic and molecular reservoirs during environmental quenching. Paper II should state which additional connection is made to the ionising-source mixture, rather than presenting the reservoir structure itself as unprecedented.

A declining line surface brightness and a rising line ratio need not imply an increase in the absolute power supplied by a hard source. They can arise when the components decline at different rates. Hot low-mass evolved stars (HOLMES), including post-AGB stars, are one candidate distributed source (Flores-Fajardo et al. 2011; Byler et al. 2019). Leaked stellar radiation is another: Belfiore et al. (2022) and the October preprint of Baron et al. (2026) give distinct, relevant treatments of diffuse excitation. The latter particularly cautions against assigning all hard, faint emission in star-forming discs to old stars.

In the model, a persistent ionising-photon production rate is not a fixed nebular luminosity floor: the gas must still intercept and absorb those photons. Conversely, a non-detection is a sensitivity statement, not a direct measure of zero gas mass. These points are essential when connecting a fading SF component to detectable NSF and undetected Balmer emission.

The density and dispersion diagnostics remain secondary tests. H II-region density estimates in the VESTIGE studies concern a different gas selection and estimator from an NSF [S II] doublet measurement. The current MAUVE saved profiles do not establish a universal increase of [S II]6716/6731 across stages, and the low-density limit limits its resolving power. Neither a universal density decline nor monotonic dynamical heating should be written as a result before those analyses are checked. Law et al. (2021), Kewley et al. (2019a), and Barnes et al. (2021) supply the appropriate diagnostic context.

Finally, the local enhancement case needs its existing history. Lizee et al. (2021) already identifies compressed gas and enhanced efficiency in NGC 4654, while Vollmer (2003) models combined gravitational and ram-pressure effects. The proposed Fourier result adds a quantified angular association; it does not establish RPS as the unique cause of the asymmetry. NGC 4424's merger, central activity, and outflow interpretation have a direct reference in Boselli et al. (2018b). Other MAUVE outflow studies further motivate distinguishing feedback from an environmental excitation signature.

A9. Terminology: should the paper use 'residual emission'?

The phrase is grammatically natural, but it is not the best primary classification for this paper. In the relevant literature, DIG, LIER emission, non-H II-like excitation, and emission associated with weak recent SF have more specific meanings. The targeted search did not establish 'residual emission' as a standard umbrella term for all of these. This does not mean the phrase is prohibited or never used.

Term	Recommended use
NSF-selected emission	Preferred label for the actual complement of the adopted SF selection, with the detection and quality requirements stated
Emission outside the adopted SF selection	Clearest wording at first use and in discussion of the census
Non-H II-like emission/excitation	Use when valid line-ratio classifications support this distinction; not automatically for every NSF-selected spaxel
LIER-like emission	Use for the low-ionisation subset supported by the diagnostic diagrams; not a synonym for all non-SF emission or proof of an AGN
Diffuse ionised gas, or DIG	Use when a spatial or emission-decomposition criterion identifies a diffuse component; DIG can be photoionised by young stars
Weak line emission	Describes intensity only; it does not identify the source or geometry
Residual ionised emission	Possible descriptive wording after defining what remains detectable, but avoid as a category name or an assumed evolutionary remnant

The main semantic problem with 'residual' is that it can suggest a subtraction residual or a remnant of an earlier phase. The data measure emission in a selected population; they do not track the same gas through time. 'Remnant H II emission' would be more problematic because it could suggest a recombining nebula persisting without a continuing energy source. A model-specific residual should be named explicitly, for example 'the emission remaining after subtraction of the modelled H II-region contribution'.

A suitable first-use definition is:

> *We divide the usable footprint into SF-selected regions, regions without the required Balmer detections, and detected regions outside the adopted SF selection. We refer to the last category as NSF-selected emission. This observational category does not assume a unique ionising source or a purely diffuse gas morphology.*

Subsequently, the paper can describe a subset as 'predominantly LIER-like' or 'consistent with a substantial DIG contribution' if the measurements support that wording. The abbreviation NSF need not assert that the true SFR is exactly zero. This distinction is especially relevant because the current selection also uses equivalent width and gas dispersion, rather than a single excitation boundary.

A10. The resulting manuscript progression

Section 3 establishes the behaviour of the SF-selected component: the nearby-survey comparison, stage-dependent conditional intensity, and the NGC 4654 local excess. Section 4 then asks how much of the observed disc retains such emission, and where lower SF coverage is accompanied by NSF-selected emission or Balmer non-detection. Section 5 measures the absolute brightness and excitation of the detected NSF component. Section 6 tests whether declining SF and a changing mixture of ionising sources can explain these connected observations, while considering gas depletion, reduced efficiency, leakage, shocks, and feedback.

The main result can be framed without an unsupported priority claim:

> *We connect the stage dependence of the intensity of SF-selected emission to changes in its spatial coverage and to the properties of the detectable emission outside that selection. This joint analysis distinguishes outer regions increasingly failing the Balmer-detection criteria from inner regions where non-SF-selected emission remains prominent, and tests how the latter changes as the young stellar contribution weakens.*

That statement describes the observational connection indicated by the current saved analyses. Its quantitative strength still depends on the comparison and selection checks documented in the earlier report. The fraction of individual regions that physically change from one state to another, a unique ionisation mechanism, and the time spent in each state remain separate questions. The Introduction below is organised to make this connection the natural next step after the existing literature.

B. Revised manuscript Section 1: Introduction

The dependence of galaxy properties on environment is a fundamental constraint on galaxy evolution. Galaxies in dense environments are more frequently associated with early-type morphologies and reduced star formation activity than those in lower-density environments, as established by the morphology-density relation and subsequent spectroscopic studies of nearby galaxy populations (Dressler 1980; Lewis et al. 2002; Gomez et al. 2003). Large surveys further show that both stellar mass and environment are associated with the probability of quenching, with environmental effects particularly relevant to satellite galaxies (Peng et al. 2010, 2012; Wetzel et al. 2013). These population trends establish the importance of environment, but do not uniquely identify how the star-forming interstellar medium (ISM) is transformed or how the decline proceeds within individual discs (see Boselli & Gavazzi 2006; Cortese, Catinella & Smith 2021; Bluck 2026, for reviews). Understanding the connection between environmental processing and quenching therefore requires spatially resolved measurements of the sites of recent star formation and of the gas that remains detectable as their contribution declines.

Several physical mechanisms can contribute to this transformation. Ram pressure exerted by the intracluster medium (ICM) can displace and remove galactic gas when the external pressure becomes sufficiently strong relative to the restoring force of the galaxy (Gunn & Gott 1972). A reduction in gas accretion can instead suppress the replenishment of the star-forming reservoir, allowing continued consumption to lower subsequent star formation (Larson, Tinsley & Caldwell 1980). Gravitational encounters and repeated tidal perturbations can also redistribute gas and alter galactic structure (Moore et al. 1996). These processes can act together and need not affect all gas phases, radii, or stellar populations simultaneously. In particular, removal of an extended atomic reservoir, compression of retained gas, and disruption or depletion of molecular material can produce different local responses during the same environmental encounter (Boselli et al. 2014; Boselli, Fossati & Sun 2022). The resulting star formation history is consequently sensitive to the initial gas distribution, the structure of the galaxy, and its interaction history.

The distinction between gas removal and star formation quenching is particularly important when interpreting integrated observations. A galaxy can retain a star-forming inner disc after losing much of its outer gas, while a lower integrated SFR can reflect either reduced activity within the remaining star-forming component or a reduction in its spatial extent. Conversely, a temporary local enhancement can occur while the galaxy's total gas reservoir is being depleted. Empirical satellite histories allow a period of continued star formation before a more rapid decline, although the inferred timescales depend on the population and the definition of infall (Wetzel et al. 2013). Spatially resolved measurements are therefore needed to determine which parts of a galaxy continue to form stars and how their activity differs from that in less affected systems.

The Virgo cluster provides an unusually detailed view of these processes. Its proximity allows the atomic, molecular, stellar, and ionised components to be studied on sub-kiloparsec scales across a wide range of environmental conditions. The VLA Imaging of Virgo Spirals in Atomic Gas survey (VIVA) revealed extended, asymmetric, and strongly truncated H I distributions (Chung et al. 2009). Combined modelling and projected phase-space analyses connect these morphologies to different interaction histories and stripping states (Vollmer 2009; Yoon et al. 2017). These classifications provide a useful basis for comparing galaxies at different stages of environmental processing, although differences in orbit, orientation, internal structure, and gravitational interactions prevent the populations from being interpreted as a unique evolutionary track.

Hα imaging has long demonstrated that the suppressed integrated star formation of Virgo spirals is often associated with truncated star-forming discs, while their inner regions can retain normal or enhanced

activity (Koopmann & Kenney 2004a, 2004b). Stellar-population measurements provide complementary constraints on when star formation declined within stripped discs (Crowl & Kenney 2008). The Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) extends this view through deep, high-resolution narrow-band imaging (Boselli et al. 2018a). Its observations establish the environmental dependence of the galaxy-integrated SFMS and Hα luminosity function, as well as the loss of individual H II regions, especially beyond the effective radius (Boselli et al. 2020, 2023a, 2023b). Thus, the spatial disappearance of SF sites is an established part of the environmental picture.

The interpretation of truncation nevertheless depends on its degree and physical context. The detailed reconstruction of NGC 4330 links a rapid decline in star formation to a radially structured stripping history (Fossati et al. 2018). Across the wider Virgo population, however, moderate truncations also occur far from the cluster centre. Morgan et al. (2024) shows that reduced gas supply can produce such truncations when a threshold for star formation is included, motivating an early starvation phase followed by stronger ram pressure. The radial extent of star formation should therefore be considered together with gas morphology and stellar history, rather than treated as a unique discriminator between gas-supply suppression and direct stripping.

The intensity of the remaining star-forming component adds an independent observational dimension. Regional luminosity, size, and density measurements in VESTIGE provide a baseline for this comparison (Boselli et al. 2025). The extension to perturbed galaxies finds lower brightest-region luminosities at fixed host stellar properties, together with changes in the composite luminosity function that do not imply a uniform fading of all regions (Boselli et al. 2026a). Furthermore, compact star-forming discs in Virgo lenticulars can exhibit approximately normal SF surface brightness despite their reduced integrated activity (Boselli et al. 2022a). These results emphasise the need to distinguish a reduction in the number or extent of SF sites from a reduction in their conditional intensity. They also require explicit attention to the quantity held fixed: the luminosity of a segmented H II region at fixed host mass is not equivalent to the local SFR surface density at fixed stellar surface density.

The molecular-gas observations offer a complementary explanation of these spatial changes. The Virgo Environment Traced in CO survey (VERTICO; Brown et al. 2021) shows that environmental effects extend beyond the atomic phase. Molecular discs become more truncated and lose low-surface-density material in H I-deficient systems, while the retained inner gas can also have lower surface densities (Zabel et al. 2022; Watts et al. 2023). At the same time, changes in the efficiency with which molecular gas forms stars contribute to suppression (Villanueva et al. 2022; Jimenez-Donaire et al. 2023). In particular, the resolved analysis of Brown et al. (2023) connects lower SFR surface density at fixed stellar surface density to both reduced molecular content and lower molecular SF efficiency. Consequently, neither the presence of detectable gas nor a familiar ensemble gas-SFR relation guarantees that the local star-forming process is unaffected by environment.

Integral-field spectroscopy (IFS) makes it possible to examine these changes through the resolved star formation main sequence (rSFMS), which relates the SFR surface density to the underlying stellar mass surface density. This relation has been measured on kiloparsec scales in CALIFA, SAMI, MaNGA, and GASP, with substantial variation associated with galaxy properties and local structure (Cano-Diaz et al. 2016; Medling et al. 2018; Ellison et al. 2018; Vulcani et al. 2019). It provides a local reference against which suppression and enhancement can be measured at comparable stellar surface density. However, a relation constructed only from SF-selected spaxels describes the component that satisfies that selection. It does not, by itself, determine how much of the surrounding disc has ceased to meet the same criteria.

Environmental IFS studies already demonstrate that both the strength and radial structure of SF can vary between populations. SAMI measurements connect dense environments to more centrally concentrated SF

(Schaefer et al. 2017). MaNGA analyses find satellite suppression across radii and stronger environmental effects in some low-mass satellite populations, while separating the properties associated with local SF intensity from those associated with quenching (Spindler et al. 2018; Lian et al. 2019; Bluck et al. 2020a, 2020b). More recently, Cakir et al. (2026) finds increasingly suppressed outer SF in asymmetric and truncated SAMI cluster galaxies, with central activity comparable to that of less perturbed systems. These studies provide important precedents for a stage-dependent analysis, while differing in spatial resolution, sample selection, and the treatment of regions with weak emission lines.

Nearby MUSE surveys now allow the rSFMS to be examined on scales approaching individual star-forming complexes. The MUSE Atlas of Disks (MAD) and PHANGS-MUSE resolve nearby discs at scales of order 100 pc or smaller (Erroz-Ferrer et al. 2019; Emsellem et al. 2022). At these scales, spatial sampling and the different lifetimes of gas and SF tracers become important (Kruijssen & Longmore 2014). Using PHANGS data, Pessa et al. (2021) demonstrates that completeness and the treatment of non-detections change the measured resolved scaling relations and their apparent scale dependence. Comparisons must therefore account for the physical resolution, averaging procedure, diffuse-emission treatment, and selected footprint. An overlap between pooled distributions need not imply that the underlying environmental populations have identical conditional SF intensities or SF coverage.

Local enhancement introduces a further dimension to this comparison. Studies of the Gas Stripping Phenomena in galaxies with MUSE survey (GASP) find elevated integrated and spatially resolved SF in selected stripping galaxies (Vulcani et al. 2018, 2020b). HST imaging also reveals clumps whose Hα luminosities at a given size exceed those of comparison regions in ordinary local star-forming galaxies (Giunchi et al. 2023). In Virgo, the compressed regions of IC 3476 and NGC 4654 provide resolved examples of enhanced activity (Boselli et al. 2021; Lizee et al. 2021). VERTICO additionally finds excess molecular gas and SF on the leading sides of H I-tailed galaxies without requiring a shorter depletion time (Roberts et al. 2023). Local SF enhancement can therefore reflect changes in available fuel, conversion efficiency, or both.

Such enhancement is not a universal property of environmentally affected galaxies. Earlier Virgo observations find no general increase in molecular SF efficiency (Vollmer et al. 2012), and recent GASP molecular measurements show normal-to-low efficiencies with strong spatial dependence (Moretti et al. 2026). A large UNIONS satellite sample shows no systematic asymmetric enhancement, while TNG50 simulations distinguish bursts along individual histories from an enhancement of the observed population as a whole (Foster et al. 2025; Goller et al. 2023). GASP also contains abruptly quenched systems and galaxies with strongly truncated gas discs, demonstrating the importance of following the population beyond the conspicuous stripping phase (Vulcani et al. 2020a; Poggianti et al. 2025). A search for local excess is therefore most informative when embedded within an analysis of the wider progression towards suppression.

As the coverage of SF-selected emission decreases, the ionised medium need not simply disappear from the observed footprint. Some locations fall below the sensitivity of the observations, whereas others retain detectable line emission with different excitation properties. Extended low-ionisation emission-line regions (LIERs) occur in both star-forming and quiescent galaxies, and can trace distributed ionising sources rather than a central active galactic nucleus (Belfiore et al. 2016). The resolved transition towards quiescence consequently involves spatially distinct combinations of SF, LIER-like, and weak or undetected emission (Belfiore et al. 2017). These established configurations motivate an environmental census that considers detectable emission outside the SF selection together with the SF component and the non-detections.

The physical interpretation of this census requires care because an excitation class is not a unique gas phase or energy source. Diffuse ionised gas (DIG) is widespread in galactic discs and can be maintained by

ionising radiation escaping from young stellar populations, with additional contributions depending on the local conditions (Haffner et al. 2009). At low Hα surface brightness, the relative strengths of low-ionisation forbidden lines commonly increase, affecting the position of regions in emission-line diagnostic diagrams (Zhang et al. 2017). The measured mixture also depends on aperture and spatial resolution (Mannucci et al. 2021). Therefore, a change in diagnostic ratios must be interpreted alongside the absolute line surface brightness and the spatial distribution of the emission.

Equivalent width adds information about the ionising output relative to the underlying stellar continuum. Weak Hα emission associated with old stellar populations can occupy regions of diagnostic diagrams commonly attributed to non-stellar ionisation in unresolved observations (Stasinska et al. 2008; Cid Fernandes et al. 2011). Resolved studies distinguish diffuse and mixed components across the Hubble sequence using equivalent width and related diagnostics (Lacerda et al. 2018). However, no single equivalent-width or surface-brightness threshold universally isolates DIG in stripped galaxies (Tomicic et al. 2021a). A measured decline in equivalent width is thus informative about the changing prominence of line emission, but does not alone identify its source.

Hot low-mass evolved stars, including post-asymptotic giant branch populations, provide a plausible distributed ionising field where recent SF is weak (Flores-Fajardo et al. 2011). Stellar-population and photoionisation calculations predict weak Hα emission and LIER-like line ratios under suitable gas conditions (Byler et al. 2019). In PHANGS-MUSE, Belfiore et al. (2022) investigates the relative contributions of leaked radiation and evolved stars to DIG. More recently, Baron et al. (2026) proposes that propagation and spectral hardening of leaked radiation can explain important sequences connecting H II regions and diffuse emission in actively star-forming discs. These alternatives make the spatial relationship to SF, the stellar population, and the available photon budget essential to interpreting harder emission in an environmentally suppressed population.

Studies of stripped galaxies provide further constraints. GASP measurements find no clear stripped-control difference in the integrated DIG Hα contribution, despite a wide range of diffuse fractions (Tomicic et al. 2021a). This concerns an emission fraction rather than an area fraction. Diffuse tails can also exhibit excess [O I]/Hα requiring additional excitation, potentially involving shocks or interactions with the hot surrounding medium (Fossati et al. 2016; Tomicic et al. 2021b). Interface-region spectroscopy connects changing excitation and SF activity to gas-disc truncation and stripping phase (Pedrini et al. 2022). These results establish that environmental excitation changes are already observed, while leaving open how they relate quantitatively to the changing coverage and brightness of emission retained within galactic discs.

Tracers sensitive to different stellar timescales provide a complementary perspective on this retained emission. Comparisons of UV and Hα in late-stage GASP galaxies associate their discrepancies in DIG-dominated regions with recently declining SF, while also identifying ionised regions with little evidence for an accompanying young UV population (Tomicic et al. 2024). In older Virgo systems, UV emission itself can have a substantial contribution from evolved populations (Martocchia et al. 2025). Such observations emphasise that the presence of a tracer is not equivalent to an ongoing SFR without an appropriate source model. The changing ionised-emission census must therefore be interpreted in conjunction with its spectral properties and stellar context.

Gas kinematics and density-sensitive line ratios provide additional constraints. Resolved MaNGA measurements connect broader gas profiles with non-H II-like excitation (Law et al. 2021), while shock models demonstrate how mechanical energy can modify optical line spectra (Allen et al. 2008). Nevertheless, an observed line width can contain unresolved velocity structure as well as local gas motions. Similarly, the [S II] doublet measures the density of the emitting component under adopted atomic and thermal conditions, with limited sensitivity in the low-density regime (Kewley et al. 2019a; Barnes et al.

2021). These measurements are most informative when considered jointly with surface brightness, equivalent width, and line ratios, rather than as unique indicators of an environmental mechanism (see also Kewley, Nicholls & Sutherland 2019).

The Multiphase Astrophysics to Unveil the Virgo Environment survey (MAUVE) was designed to connect the stellar and gaseous components of Virgo galaxies through spatially resolved observations (Catinella et al. 2025). In Paper I, we used MAUVE-MUSE observations to examine the local relation between metallicity and SF, and developed a resolved gas-regulator interpretation of the competition between SF-driven and gas-supply-driven variability (Huang et al. 2026). Complementary MAUVE work already finds environmental differences in ionised-gas excitation and kinematics, interpreting the increased prominence of DIG primarily in terms of suppressed SF, with a secondary contribution from more strongly disturbed gas (Brown et al. 2026). The next question is how this changing prominence is connected to the intensity and coverage of SF across the different environmental-stage populations.

In this work, we investigate that connection using the MAUVE-MUSE observations at physical resolutions of order 100 pc. We first compare the resolved SF distribution with nearby MUSE measurements and examine the conditional intensity of SF in the adopted pre-peak, close-to-peak, and post-peak populations, including the possibility of local excess. We then measure the fractions of the usable footprint classified as SF, non-detected in the required Balmer lines (ND), or detected but outside the adopted SF selection (NSF). This division allows us to distinguish the spatial loss of SF-selected emission from changes in its intensity, and to determine where detectable NSF emission accompanies lower SF coverage. Finally, we examine the absolute line surface brightness, diagnostic ratios, equivalent width, and supplementary gas properties of the NSF-selected component. A restricted gas-supply framework is used to investigate whether the combined behaviour is consistent with declining young-star emission and a changing mixture of ionising sources. Our aim is to establish the spatial connection among these observables, while distinguishing the measured population trends from a unique history for individual gas regions.

This paper is organised as follows. Section 2 describes the observations, derived quantities, environmental-stage assignments, emission classifications, and comparison methodology. Section 3 presents the resolved SF distributions, their stage dependence, and the local excess analysis. Section 4 examines the spatial census of SF, NSF, and ND regions. Section 5 characterises the detectable emission outside the SF selection. Section 6 develops the physical interpretation, compares it with previous work, and discusses the limitations of the observations and model. Section 7 presents our conclusions.

H. Expanded annotated bibliography

The bibliography is grouped by scientific role. It includes the sources used in Parts A and B and additional, directly relevant material for the existing discussion plan. The annotations identify the purpose and limits of each reference; inclusion does not mean that every paper belongs in the opening paragraphs. Published years are used where established, rather than automatically using an earlier arXiv submission year. Author manuscripts provide accessible primary sources when a publisher page is unavailable. The October 2026 Baron et al. study remains explicitly identified as a preprint.

H1. Environment, quenching, and the broad physical framework

R001. Bluck (2026). *A review of galaxy quenching - Part I: Defining the problem and observational results.* Publication record. Published September 2026 review of the population and resolved quenching literature. Used for orientation, with primary studies cited for specific results.

R002. Boselli & Gavazzi (2006). *Environmental Effects on Late-Type Galaxies in Nearby Clusters.* Publication record; Primary manuscript. Multiphase observational and theoretical environmental review, especially Virgo, Coma, and A1367.

R003. Boselli, Fossati & Sun (2022). *Ram Pressure Stripping in High-Density Environments.* Publication record; Primary manuscript. Comprehensive RPS review; useful for the physical context, gas-phase dependence, and case-study literature.

R004. Cortese, Catinella & Smith (2021). *The Dawes Review 9: The role of cold gas stripping on the star formation quenching of satellite galaxies.* Publication record; Primary manuscript. Distinguishes cold-gas stripping from complete SF quenching and reviews the relevant timescales and observational selections.

R005. Dressler (1980). *Galaxy morphology in rich clusters: implications for the formation and evolution of galaxies.* Publication record; Primary manuscript. Foundational morphology-density observation; not a unique proof of ram-pressure-driven transformation.

R006. Gomez et al. (2003). *Galaxy Star-Formation as a Function of Environment in the Early Data Release of the Sloan Digital Sky Survey.* Publication record; Primary manuscript. SDSS environmental SF measurements; an independent population-level basis for the introductory argument.

R007. Gunn & Gott (1972). *On the Infall of Matter Into Clusters of Galaxies and Some Effects on Their Evolution.* Publication record. Foundational ram-pressure argument; later work is required for multiphase, time-dependent predictions.

R008. Larson, Tinsley & Caldwell (1980). *The evolution of disk galaxies and the origin of S0 galaxies.* Publication record; Primary manuscript. Foundational reservoir-replenishment and consumption argument; supports the starvation context.

R009. Lewis et al. (2002). *The 2dF Galaxy Redshift Survey: The environmental dependence of galaxy star formation rates near clusters.* Publication record; Primary manuscript. Population evidence for environmental suppression of SF beyond cluster cores; supports the opening motivation rather than a specific gas-removal mechanism.

R010. Moore et al. (1996). *Galaxy Harassment and the Evolution of Clusters of Galaxies.* Publication record; Primary manuscript. Gravitational harassment as an additional environmental mechanism; relevant to mixed histories, not an alternative label for hydrodynamic stripping.

R011. Peng et al. (2010). *Mass and environment as drivers of galaxy evolution in SDSS and zCOSMOS and the origin of the Schechter function.* Publication record; Primary manuscript. Phenomenological separation of mass and environmental quenching; does not uniquely identify the underlying gas physics.

R012. Peng et al. (2012). *Mass and Environment as Drivers of Galaxy Evolution II: The quenching of satellite galaxies as the origin of environmental effects.* Publication record; Primary manuscript. Satellite quenching as a major environmental contribution; retain the distinction between empirical population decomposition and mechanism.

R013. Wetzel et al. (2013). *Galaxy evolution in groups and clusters: satellite star formation histories and quenching timescales in a hierarchical Universe.* Publication record; Primary manuscript. Delayed-then-rapid satellite histories with preprocessing; inferred timescales are population- and model-dependent.

H2. Virgo histories, VESTIGE, and H II-region properties

R014. Boselli et al. (2018a). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).I. Introduction to the Survey.* Publication record; Primary manuscript. VESTIGE I: survey design, depth, and spatial resolution. Imaging morphology and spectroscopy supply complementary classifications.

R015. Boselli et al. (2018b). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).IV. A tail of Ionised Gas in the Merger Remnant NGC 4424.* Publication record; Primary manuscript. VESTIGE IV: NGC 4424's merger history, stripped gas, and central outflow interpretation; prevents a uniquely RPS-driven interpretation of its central excess.

R016. Boselli et al. (2020). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).VI. Environmental quenching on H II region scales.* Publication record; Primary manuscript. VESTIGE VI: fewer outer H II regions in stripped galaxies. This regional-abundance result is not itself a conditional-intensity measurement.

R017. Boselli et al. (2021). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).IX. The effects of ram pressure stripping down to the scale of individual H II regions in the dwarf galaxy IC 3476.* Publication record; Primary manuscript. VESTIGE IX: compressed gas and enhanced regional SF in IC 3476; a direct counterexample to universal local suppression.

R018. Boselli et al. (2022a). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).XII. Ionised gas emission in the inner regions of lenticular galaxies.* Publication record; Primary manuscript. VESTIGE XII: compact SF discs and shock-ionised filaments in lenticulars; low total SFR can coexist with retained SF surface brightness.

R019. Boselli et al. (2023a). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).XIV. The main sequence relation in a rich environment down to stellar masses of approximately 10^6 solar masses.* Publication record; Primary manuscript. VESTIGE XIV: global SFMS scatter linked to H I content. Its rejection of specific mild models is not a general theorem excluding starvation.

R020. Boselli et al. (2023b). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XV. The Hα luminosity function of the Virgo cluster.* Publication record; Primary manuscript. VESTIGE XV: galaxy Hα luminosity function. Distinguish this galaxy population statistic from the H II-region luminosity function.

R021. Boselli et al. (2025). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XVII. Statistical properties of individual H II regions in unperturbed systems.* Publication record; Primary manuscript. VESTIGE XVII: regional luminosity, size, and density baseline for H I-normal galaxies; source for consistent catalogue definitions.

R022. Boselli et al. (2026a). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). XXI. Statistical properties of individual H II regions in perturbed galaxies.* Publication record; Primary manuscript. VESTIGE XXI: the essential environmental H II-luminosity comparison for Section 3.

R023. Boselli et al. (2026b). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XX. Star formation in the tidal tail of NGC 4254.* Publication record; Primary manuscript. VESTIGE XX: molecular clouds and SF in NGC 4254's tidal tail. Relevant to stripped-gas fate, with a gravitational origin explicitly distinguished.

R024. Chung et al. (2009). *VIVA, VLA Imaging of Virgo spirals in Atomic gas: I. The Atlas & The H I Properties.* Publication record; Primary manuscript. VIVA H I morphology and spatial gas disturbances. The linked journal DOI is the original atlas, not the subsequent erratum.

R025. Crowl & Kenney (2008). *The Stellar Populations of Stripped Spiral Galaxies in the Virgo Cluster.* Publication record; Primary manuscript. Stellar populations constrain the recent SF histories of stripped Virgo discs; provides information beyond instantaneous emission-line maps.

R026. Fossati et al. (2018). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). II. Constraining the quenching time in the stripped galaxy NGC 4330.* Publication record; Primary manuscript. VESTIGE II: a radially resolved quenching history for NGC 4330, using combined imaging and spectroscopy.

R027. Koopmann & Kenney (2004a). *Massive Star Formation Rates and Radial Distributions from Hα Imaging of 84 Virgo Cluster and Isolated Spiral Galaxies.* Publication record; Primary manuscript. Radial Hα measurements distinguish truncated outer SF from retained or enhanced inner activity.

R028. Koopmann & Kenney (2004b). *Hα Morphologies and Environmental Effects in Virgo Cluster Spiral Galaxies.* Publication record; Primary manuscript. Morphological classes of Virgo Hα emission; useful for distinguishing truncation, anemic SF, and local enhancement.

R029. Martocchia et al. (2025). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XVIII. Reconstructing the star formation history of early-type galaxies through the combination of their UV and Hα emission.* Publication record; Primary manuscript. VESTIGE XVIII: evolved populations contribute to UV emission in quiescent Virgo galaxies; UV is not universally a young-star tracer.

R030. Morgan et al. (2024). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE): XVI. The ubiquity of truncated star-forming disks across the Virgo cluster environment.* Publication record; Primary manuscript. VESTIGE XVI: widespread truncation and a threshold-dependent starvation model. Essential qualification to treating all truncation as direct stripping.

R031. Vollmer (2009). *A holistic view on ram pressure stripping in the Virgo cluster - The first complete model-based time sequence.* Publication record; Primary manuscript. Model-based Virgo stripping sequence; a precedent for interpreting stage populations with explicit dynamical assumptions.

R032. Yoon et al. (2017). *A History of H I Stripping in Virgo: A Phase-space View of VIVA Galaxies.* Publication record; Primary manuscript. H I morphology combined with phase space to constrain stripping history; classes do not uniquely specify time since peak ram pressure.

H3. Molecular gas, VERTICO, and the gas-SF connection

R033. Boselli et al. (2014). *Cold gas properties of the Herschel Reference Survey. III. Molecular gas stripping in cluster galaxies.* Publication record; Primary manuscript. Herschel Reference Survey molecular-gas stripping evidence; qualifies any model assuming no direct environmental change to molecular gas.

R034. Brown et al. (2021). *VERTICO: The Virgo Environment Traced In CO Survey.* Publication record; Primary manuscript. VERTICO survey definition and CO data; anchors the molecular-gas comparison sample.

R035. Brown et al. (2023). *VERTICO VII: Environmental quenching caused by suppression of molecular gas content and star formation efficiency in Virgo Cluster galaxies.* Publication record; Primary manuscript. VERTICO VII: resolved SF suppression decomposed into molecular-content and efficiency contributions at 720 pc.

R036. Jimenez-Donaire et al. (2023). *VERTICO III: The Kennicutt-Schmidt relation in Virgo cluster galaxies.* Publication record; Primary manuscript. VERTICO III: ensemble KS relation, galaxy-to-galaxy scatter, and efficiency differences with H I deficiency.

R037. Leroy et al. (2013). *Molecular Gas and Star Formation in Nearby Disk Galaxies.* Publication record; Primary manuscript. Nearby-disc molecular gas and SF relation; establishes the depletion-time reference and its environmental variations.

R038. Lilly et al. (2013). *Gas-regulation of galaxies: the evolution of the cosmic sSFR, the metallicity-mass-SFR relation and the stellar content of haloes.* Publication record; Primary manuscript. Gas-regulator framework; model architecture is established, while application to joint stage-dependent emission remains the proposed extension.

R039. Moretti et al. (2026). *Molecular gas and star formation in GASP jellyfish galaxies.* Publication record; Primary manuscript. GASP molecular gas: normal-to-low efficiencies depend on location and density; bright clumps need not represent a uniformly efficient disc.

R040. Roberts et al. (2023). *VERTICO VI: Cold-gas asymmetries in Virgo cluster galaxies.* Publication record; Primary manuscript. VERTICO VI: leading-side molecular-gas and SF excess without an average depletion-time change; separates intensity from efficiency.

R041. Villanueva et al. (2022). *VERTICO IV: Environmental Effects on the Gas Distribution and Star Formation Efficiency of Virgo Cluster Spirals.* Publication record; Primary manuscript. VERTICO IV: reduced inner molecular SF efficiency with H I disturbance; particularly relevant to retained gas in the inner disc.

R042. Vollmer et al. (2012). *The influence of the cluster environment on the star formation efficiency of 12 Virgo spiral galaxies.* Publication record; Primary manuscript. Twelve Virgo galaxies demonstrate diverse efficiency responses; no universal windward molecular-efficiency enhancement.

R043. Watts et al. (2023). *VERTICO V: The environmentally driven evolution of the inner cold gas discs of Virgo cluster galaxies.* Publication record; Primary manuscript. VERTICO V: changes in retained inner atomic and molecular surface densities; motivates physical changes beyond outer-disc removal.

R044. Zabel et al. (2022). *VERTICO II: effects of H I-identified environmental mechanisms on molecular gas.* Publication record; Primary manuscript. VERTICO II: environmental truncation of molecular profiles and preferential loss of low-surface-density material.

H4. Resolved SF relations, completeness, and environmental IFS studies

R045. Bluck et al. (2020a). *Are galactic star formation and quenching governed by local, global or environmental phenomena?.* Publication record; Primary manuscript. Distinguishes predictors of local SF intensity from quenching; uses continuum information where emission-line SFRs are unavailable.

R046. Bluck et al. (2020b). *How do central and satellite galaxies quench? -- Insights from spatially resolved spectroscopy in the MaNGA survey.* Publication record; Primary manuscript. MaNGA radial quenching and satellite-mass dependence; a key outside-in environmental precedent.

R047. Cakir et al. (2026). *The SAMI Galaxy Survey: Quenching of Star Formation in Clusters III. Ram-Pressure-Affected Galaxy Populations.* Publication record; Primary manuscript. SAMI asymmetric/truncated cluster populations show stronger outer suppression without central enhancement; a close stage-comparison precedent.

R048. Cano-Diaz et al. (2016). *Spatially-Resolved Star Formation Main Sequence of Galaxies in the CALIFA Survey.* Publication record; Primary manuscript. CALIFA rSFMS at kiloparsec scales; a foundational resolved-scaling precedent.

R049. Ellison et al. (2018). *Star formation is boosted (and quenched) from the inside out: radial star formation profiles from MaNGA.* Publication record; Primary manuscript. MaNGA radial enhancement and suppression relative to the rSFMS; radial shape is not unique to cluster stripping.

R050. Emsellem et al. (2022). *The PHANGS-MUSE survey -- Probing the chemo-dynamical evolution of disc galaxies.* Publication record; Primary manuscript. PHANGS-MUSE survey design, spatial resolution, and sample selection; do not assume a strictly field-only control.

R051. Erroz-Ferrer et al. (2019). *The MUSE Atlas of Disks (MAD): Resolving Star Formation Rates and Gas Metallicities on < 100pc Scales.* Publication record; Primary manuscript. MAD measurements below approximately 100 pc; demonstrates relevant high-resolution MUSE work beyond PHANGS.

R052. Kennicutt & Evans (2012). *Star Formation in the Milky Way and Nearby Galaxies.* Publication record; Primary manuscript. SFR tracer calibrations and temporal sensitivity; supports explicit IMF, timescale, and attenuation conventions.

R053. Kruijssen & Longmore (2014). *An uncertainty principle for star formation. I. Why galactic star formation relations break down below a certain spatial scale.* Publication record; Primary manuscript. Spatial sampling of short-lived SF phases limits small-scale scaling-relation interpretation.

R054. Lian et al. (2019). *SDSS-IV MaNGA: Environmental dependence of gas metallicity gradients in local star-forming galaxies.* Publication record; Primary manuscript. MaNGA low-mass satellites show environment-dependent SF profiles. Used here for SF results, not to expand Paper II into metallicity analysis.

R055. Medling et al. (2018). *The SAMI Galaxy Survey: Spatially Resolving the Main Sequence of Star Formation.* Publication record; Primary manuscript. SAMI resolved main sequence; establishes internal and environmental variation around a local SF relation.

R056. Pessa et al. (2021). *Star formation scaling relations at approximately 100 pc from PHANGS: Impact of completeness and spatial scale.* Publication record; Primary manuscript. Direct PHANGS comparison for completeness, scale, and non-detection treatment; align the estimator before interpreting normalisations.

R057. Santoro et al. (2022). *PHANGS-MUSE: the H II region luminosity function of local star-forming galaxies.* Publication record; Primary manuscript. H II-region luminosity functions depend on SF and internal environment; a comparison for regional statistics rather than a spaxel intensity estimator.

R058. Schaefer et al. (2017). *The SAMI Galaxy Survey: Spatially resolving the environmental quenching of star formation in GAMA galaxies.* Publication record; Primary manuscript. SAMI environmental variation of SF concentration; relevant to radial suppression.

R059. Spindler et al. (2018). *SDSS-IV MaNGA: The Spatial Distribution of Star Formation and its Dependence on Mass, Structure and Environment.* Publication record; Primary manuscript. MaNGA satellite suppression across radii. Uses Hα and continuum-based estimates, unlike an exclusively Balmer-detected census.

R060. Vulcani et al. (2019). *GASP. XX. From the loose spatially-resolved to the tight global SFR-Mass relation in local spiral galaxies.* Publication record; Primary manuscript. GASP XX: galaxy-to-galaxy variation and local structure broaden the resolved SF relation in undisturbed discs.

H5. Local enhancement, late-stage suppression, and simulation comparisons

R061. Foster et al. (2025). *No Evidence of Asymmetrically Enhanced Star Formation in Infalling Galaxies in UNIONS.* Publication record; Primary manuscript. UNIONS satellite population shows no systematic asymmetric SF enhancement; an important observational counterweight.

R062. Giunchi et al. (2023). *HST imaging of star-forming clumps in 6 GASP ram-pressure stripped galaxies.* Publication record; Primary manuscript. HST clumps in six strongly stripped GASP galaxies; luminosity at fixed size supplies a direct regional-intensity comparison.

R063. Goller et al. (2023). *Jellyfish galaxies with the IllustrisTNG simulations -- No enhanced population-wide star formation according to TNG50.* Publication record; Primary manuscript. TNG50 distinguishes individual bursts from population-wide enhancement. Simulation outcome, not a direct measurement of MAUVE galaxies.

R064. Lizee et al. (2021). *Gas compression and stellar feedback in the tidally interacting and ram-pressure stripped Virgo spiral galaxy NGC 4654.* Publication record; Primary manuscript. NGC 4654 compression and elevated local efficiency already established; the MAUVE angular analysis is a quantitative extension.

R065. Poggianti et al. (2025). *The MUSE view of ram pressure stripped galaxies in clusters: the GASP sample.* Publication record; Primary manuscript. Full GASP cluster sample, stripping classes, truncated discs, and selection limitations; essential for placing individual results in context.

R066. Roberts & Parker (2020). *Ram pressure stripping candidates in the Coma Cluster: Evidence for enhanced star formation.* Publication record; Primary manuscript. Coma stripping candidates show enhanced SF; another selected-population comparison beyond Virgo and GASP.

R067. Steinhauser et al. (2016). *Simulations of ram-pressure stripping in galaxy-cluster interactions.* Publication record; Primary manuscript. Simulated galaxy-cluster RPS encounters; responses depend on physical and orbital conditions.

R068. Tonnesen & Bryan (2012). *Star Formation in Ram Pressure Stripped Tails.* Publication record; Primary manuscript. Simulated star formation in stripped tails; tail conditions should not be transferred directly to retained inner-disc NSF gas.

R069. Vollmer (2003). *NGC 4654: gravitational interaction or ram pressure stripping?.* Publication record; Primary manuscript. Combined tidal and ram-pressure model for NGC 4654; an angular association alone does not remove the tidal alternative.

R070. Vulcani et al. (2018). *Enhanced star formation in both disks and ram pressure stripped tails of GASP jellyfish galaxies.* Publication record; Primary manuscript. Integrated enhancement in selected GASP stripping galaxies; selection and stage coverage matter in comparisons.

R071. Vulcani et al. (2020a). *GASP XXIV. The history of abruptly quenched galaxies in clusters.* Publication record; Primary manuscript. GASP XXIV: abruptly quenched galaxies with largely outside-in histories; supplies the later-stage counterpart to enhancement studies.

R072. Vulcani et al. (2020b). *GASP XXX. The spatially resolved SFR-Mass relation in stripping galaxies in the local universe.* Publication record; Primary manuscript. GASP XXX: resolved SF enhancement at roughly kiloparsec scales; distinguish this from the more inclusive MAUVE stage population.

R073. Vulcani et al. (2024). *Evidence for enhanced star formation rates in zapproximately 0.35 cluster galaxies undergoing ram pressure stripping.* Publication record; Primary manuscript. Enhancement among selected stripped galaxies at intermediate redshift; supports environmental diversity without serving as a matched local control.

R074. Zhu, Tonnesen & Bryan (2024). *When and how does ram pressure stripping in low-mass satellite galaxies enhance star formation.* Publication record; Primary manuscript. Models how ram pressure can enhance SF in low-mass satellites; relevant to mechanism and temporal dependence.

H6. DIG, LIER emission, and excitation in stripped systems

R075. Baron et al. (2026). *DIGging into H II regions: the role of ionizing photon leakage and spectral hardening in coupling H II regions and diffuse ionized gas.* Primary manuscript. October preprint: leaked radiation and spectral hardening link H II and diffuse emission. Relevant alternative, chiefly tested in actively star-forming discs.

R076. Belfiore et al. (2016). *SDSS IV MaNGA - Spatially resolved diagnostic diagrams: A proof that many galaxies are LIERs.* Publication record; Primary manuscript. MaNGA extended LIER emission and distributed-source evidence; provides the appropriate terminology and a direct observational precedent.

R077. Belfiore et al. (2017). *SDSS IV MaNGA - The spatially resolved transition from star formation to quiescence.* Publication record; Primary manuscript. Resolved transition towards quiescence; central LIER configurations are not by themselves a cluster stripping sequence.

R078. Belfiore et al. (2022). *A tale of two DIGs: The relative role of H II regions and low-mass hot evolved stars in powering the diffuse ionised gas (DIG) in PHANGS-MUSE galaxies.* Publication record; Primary manuscript. PHANGS-MUSE leakage and evolved-star contributions; supports a mixed ionising-source interpretation and photon-budget tests.

R079. Byler et al. (2019). *Self-consistent predictions for LIER-like emission lines from post-AGB stars.* Publication record; Primary manuscript. Post-AGB photoionisation predictions for weak-EW, LIER-like emission; nebular luminosity still depends on absorbing gas.

R080. Cid Fernandes et al. (2011). *A comprehensive classification of galaxies in the SDSS: How to tell true from fake AGN?.* Publication record; Primary manuscript. WHAN classification and weak-line galaxies; integrated thresholds need explicit justification when applied to resolved spaxels.

R081. Flores-Fajardo et al. (2011). *Ionization of the diffuse gas in galaxies: Hot low-mass evolved stars at work.* Publication record; Primary manuscript. HOLMES as distributed ionising sources; a physical candidate, not an automatic identification of every NSF location.

R082. Fossati et al. (2016). *MUSE sneaks a peek at extreme ram-pressure stripping events. II. The physical properties of the gas tail of ESO137-001.* Publication record; Primary manuscript. Physical and excitation properties of ESO 137-001's tail; spatially separates compact SF and diffuse gas.

R083. Fossati et al. (2019). *MUSE sneaks a peek at extreme ram-pressure stripping events - IV. Hydrodynamic and gravitational interactions in the Blue Infalling Group.* Publication record; Primary manuscript. Blue Infalling Group: gravitational and hydrodynamic effects with mixed photoionisation and shocks; not proof of HOLMES in inner Virgo discs.

R084. Fumagalli et al. (2014). *MUSE sneaks a peek at extreme ram-pressure stripping events. I. A kinematic study of the archetypal galaxy ESO137-001.* Publication record; Primary manuscript. ESO 137-001 gas kinematics; a case-study basis for interpreting stripped-tail dynamics.

R085. Haffner et al. (2009). *The warm ionized medium in spiral galaxies.* Publication record; Primary manuscript. Warm ionised medium review; DIG describes a distributed gas component and does not imply an exclusively old ionising population.

R086. Lacerda et al. (2018). *Diffuse ionized gas in galaxies across the Hubble sequence at the CALIFA resolution.* Publication record; Primary manuscript. CALIFA equivalent-width distributions distinguish diffuse and mixed emission; useful context rather than a universal physical classifier.

R087. Luo et al. (2023). *Tracing the kinematics of the whole ram pressure stripped tails in ESO 137-001.* Publication record; Primary manuscript. Extended ESO 137-001 tail kinematics; useful for the limits of attributing every line-width increase to local turbulence.

R088. Mannucci et al. (2021). *The diffuse ionized gas (DIG) in star-forming galaxies: the influence of aperture effects on local H II regions.* Publication record; Primary manuscript. Aperture effects in H II-region/DIG comparisons; prevents conflating an emission mixture with a uniquely different physical population.

R089. Pedrini et al. (2022). *MUSE sneaks a peek at extreme ram-pressure stripping events -- V. Towards a complete view of the galaxy cluster A1367.* Publication record; Primary manuscript. Interface excitation linked to truncation and stripping phase; a direct precedent for connecting stage and ionisation.

R090. Poggianti et al. (2019). *GASP XIII. Star formation in gas outside galaxies.* Publication record; Primary manuscript. GASP XIII: physical properties of SF outside galaxies; compact tail clumps are not a census of all diffuse gas.

R091. Stasinska et al. (2008). *Can retired galaxies mimic active galaxies? Clues from the Sloan Digital Sky Survey.* Publication record; Primary manuscript. Retired stellar populations can produce apparent AGN-like diagnostic ratios; source interpretation requires more than a line-ratio location.

R092. Tomicic et al. (2021a). *GASP XXXII. Measuring the diffuse ionized gas fraction in ram-pressure stripped galaxies.* Publication record; Primary manuscript. GASP XXXII: DIG flux-fraction estimator and stripped-control comparison; the denominator differs from NSF area coverage.

R093. Tomicic et al. (2021b). *GASP XXXV: Characteristics of the diffuse ionised gas in gas-stripped galaxies.* Publication record; Primary manuscript. GASP XXXV: diffuse-tail excitation and additional processes; the tail geometry must be distinguished from the retained inner disc.

R094. Tomicic et al. (2024). *Spatially resolved comparison of SFRs from UV and Hα in GASP gas-stripped galaxies.* Publication record; Primary manuscript. UV/Hα discrepancies in late-stage GASP systems provide evidence relevant to declining SF and non-SF ionisation.

R095. Zhang et al. (2017). *SDSS-IV MaNGA: The Impact of Diffuse Ionized Gas on Emission-line Ratios, Interpretation of Diagnostic Diagrams, and Gas Metallicity Measurements.* Publication record; Primary manuscript. DIG surface brightness and line-ratio changes; necessary context for weaker absolute lines with higher forbidden-line/Balmer ratios.

H7. Line diagnostics, density, and kinematic interpretation

R096. Allen et al. (2008). *The MAPPINGS III Library of Fast Radiative Shock Models.* Publication record; Primary manuscript. Radiative-shock model library; use matched line ratios and model conditions rather than a broad-line label alone.

R097. Barnes et al. (2021). *Comparing the pre-SNe feedback and environmental pressures for 6000 H II regions across 19 nearby spiral galaxies.* Publication record; Primary manuscript. H II-region feedback and environmental pressures; resolved regional densities are differently weighted from NSF doublet measurements.

R098. Kewley et al. (2001). *Theoretical Modeling of Starburst Galaxies.* Publication record; Primary manuscript. Theoretical starburst boundaries in diagnostic diagrams; classification boundaries do not identify a unique non-stellar source.

R099. Kewley et al. (2006). *The Host Galaxies and Classification of Active Galactic Nuclei.* Publication record; Primary manuscript. Seyfert/LINER diagnostic distinctions; later resolved LIER work modifies the interpretation of spatially extended low-ionisation emission.

R100. Kewley et al. (2019a). *Theoretical ISM pressure and electron density diagnostics for local and high-redshift galaxies.* Publication record; Primary manuscript. Density and pressure diagnostics, including assumptions and limiting regimes relevant to the [S II] doublet.

R101. Kewley, Nicholls & Sutherland (2019). *Understanding Galaxy Evolution through Emission Lines.* Publication record; Primary manuscript. Emission-line physics review; joint interpretation of ionisation, abundance, density, and pressure diagnostics.

R102. Law et al. (2021). *SDSS-IV MaNGA: Refining Strong Line Diagnostic Classifications Using Spatially Resolved Gas Dynamics.* Publication record; Primary manuscript. MaNGA excitation and gas-dispersion relations; relevant to independent kinematic checks and selection effects.

R103. Luridiana et al. (2015). *PyNeb: a new tool for analyzing emission lines. I. Code description and validation of results.* Publication record; Primary manuscript. PyNeb atomic-emission calculations; a methods reference for the eventual density analysis, not evidence for an environmental density trend.

R104. Sanchez et al. (2024). *The WHaD diagram: Classifying the ionizing source with one single emission line.* Publication record; Primary manuscript. WHaD classification uses Hα equivalent width and dispersion; a selected kinematic difference is not automatically independent evidence.

R105. Sanders et al. (2016). *The MOSDEF Survey: Electron Density and Ionization Parameter at z approximately 2.3.* Publication record; Primary manuscript. Electron-density and ionisation-parameter analysis. Methods comparison only; its high-redshift sample is not a Virgo evolutionary analogue.

H8. MAUVE context, Paper I, and local feedback

R106. Attwater et al. (2026). *MAUVE-MUSE: A Star Formation-driven Outflow Caught in the Act of Quenching the Stripped Virgo Galaxy NGC 4064.* Publication record; Primary manuscript. NGC 4064 star-formation-driven outflow in a stripped Virgo galaxy; local feedback can accompany environmental suppression.

R107. Brown et al. (2026). *MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.* Publication record; Primary manuscript. Existing MAUVE excitation and kinematic interpretation; the strongest constraint on claims that suppressed SF first reveals DIG in Paper II.

R108. Catinella et al. (2025). *Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE).* Publication record. MAUVE survey design and scientific motivation; the analysed Paper II product groups should be reported separately from the survey design sample.

R109. Cortese et al. (2026). *MAUVE: Cold neutral gas in the outflow of NGC 4383 and evidence for a fountain flow.* Publication record; Primary manuscript. NGC 4383 neutral outflow/fountain constraints; illustrates the multiphase information needed to infer gas fate.

R110. Huang et al. (2026). *MAUVE-MUSE: When Metallicity Follows or Fights Star Formation -- A Mass-Dependent Inversion in Virgo Galaxies.* Publication record; Primary manuscript. Paper I: style reference and resolved regulator framework. Paper II retains a distinct environmental and emission-census objective.

R111. Watts et al. (2024). *MAUVE: a 6 kpc bipolar outflow launched from NGC 4383, one of the most H I-rich galaxies in the Virgo cluster.* Publication record. MAUVE outflow precedent; feedback can affect ionised-gas line widths and excitation without uniquely tracing ram pressure.